DEEP RESEARCH Prompt:

Scientist Information

- **Name:** Michael Wigler
- **Dates:** 1947 - Present
- **Era/Field:** Cancer Biology & Oncogenes
- **Course context:** BIOL 3030 intro biochemistry audience; collaborative class eBook; final narrative will be ~500 words written by the student

Your Mission

Collect high-signal raw materials (facts, quotes, scenes, human details) that will allow me to write a short narrative profile. Do not choose or assume a narrative lens at the start. First do unbiased research. Only after compiling evidence, recommend the best narrative lens using the scoring rubric at the end.

Core Constraints (Non-Negotiable)

- **No finished prose profile.** Output research materials only.
- **Citations for every meaningful claim.** If you cannot cite it, label it as unverified and treat it as suspect.
- **Prefer primary sources.** Letters, interviews, notebooks, Nobel lectures, oral histories, autobiographies.
- **Myth defense:** Flag details that are widely repeated but weakly sourced (especially precise timestamps, dramatic dialogue, too-perfect anecdotes).
- **Accuracy over drama:** Vividness only counts if it is documented.

What You Need to Find

1) Anchor Scene Candidates (Find 3–6)

Identify multiple specific events that could plausibly serve as the backbone of a narrative. For each scene, capture:

- **What happened:** discovery, decision, conflict, setback, collaboration, turning point, institutional barrier
- **When and where:** date (or best estimate), institution/lab/city, conference, wartime context, etc.
- **Who was involved:** key people present or affected
- **Concrete texture:** workspace, instrument/method, letter content, meeting setting, observed behavior, reactions
- **Why it mattered:** scientific impact, career impact, human impact
- **Evidence quality:** primary vs secondary; how direct the documentation is
- **Myth risk:** is this a “too perfect” story that may be repeated without strong sourcing?

Output each scene as a card:

SCENE #1: [Short title]

- Summary:
- Date/Place:
- People:
- Concrete details (quote-level if available):
- Why it matters:
- Evidence quality (primary/secondary; strength):
- Myth risk (low/medium/high + why):
- Sources:

(Repeat for 3–6 scenes.)

2) Direct Quotes From the Scientist (Target 3–7)

Find quotes said or written by the scientist (not quotes about them). Prioritize quotes that reveal how they think or who they are.

For each quote:

- Exact quote
- Context/date (letter, interview, lecture, notebook, memoir)
- Where it appears (archive, transcript, book, Nobel lecture page)
- Full citation

3) Human Reality Details (Collect 8–15, Source-Backed)

Not trivia for trivia's sake. Details that help a reader see the person and connect to how they worked or thought.

Look for:

- upbringing and constraints, formative experiences, obstacles
- habits, quirks, working style, mentoring style, lab culture
- colleagues' or students' recollections (with careful sourcing)
- personal life only if documented and clearly relevant (avoid gossip)

Format:

- Detail (1 sentence) — Source:
- Detail — Source:

4) Science Connection to Intro Biochemistry (BIOL 3030 Bridge)

Identify:

- the best-fitting intro biochemistry topic this scientist connects to (enzymes, metabolism, DNA/RNA, protein structure, membranes, signaling, methods)
- their precise contribution (1 sentence, accurate and specific)
- one teachable bridge explanation for intro students (2–4 factual sentences)

- if available, an analogy or explanation the scientist themselves used (with citation)

5) Complications, Disputes, or Controversies (Only if Evidence Exists)

Without sensationalizing, look for:

- priority disputes, credit assignment, ethics, politics, institutional barriers
- later corrections or re-interpretations of their work
- documented criticism from peers, or their criticism of others

For each:

- What's agreed upon:
- What's disputed:
- Best sources on both sides (if applicable):

6) "Not on Wikipedia" Gold (Aim for 5+ items)

Hunt in:

- journal memorials/obituaries (Nature, Science, PNAS, society memoirs)
- oral histories (CSHL, NIH, National Academies, university archives)
- correspondence collections, lab notebooks, archival finding aids
- scholarly biographies or history-of-science chapters

For each item:

- Finding — Why it is distinctive — Source:

Required Sources

Minimum 6 sources, preferably 10+. Prioritize:

1. Primary sources
2. Peer-reviewed memorials/academy memoirs
3. Scholarly biographies/history-of-science articles/books (reputable presses)
4. Institutional archives and museum collections
5. Careful secondary sources only when necessary

Output Format (Strict)

A) ANCHOR SCENES (3–6)

SCENE #1: The Birth of Gene Transfer

- **Summary:** Developing the first reliable method for introducing foreign DNA into mammalian cells, turning them into "protein-producing machines."
- **Date/Place:** May 1977, Columbia University, New York City.
- **People:** Michael Wigler, Richard Axel, Saul Silverstein.
- **Concrete details:** Working as a PhD student in Silverstein's lab, Wigler used a calcium phosphate precipitate to "trick" cells into absorbing the *tk* gene from the Herpes Simplex Virus. This "cotransformation" technique allowed scientists to insert any gene into animal cells by pairing it with a selectable marker.
- **Why it matters:** This tool laid the groundwork for the entire biotech industry and earned Columbia University nearly $800 million in patent royalties. It allowed for the production of medicinal proteins like erythropoietin.
- **Evidence quality:** High. Primary publication in *Cell* (1977) and extensive documentation in NIH patent case studies.
- **Myth risk:** Low. The technical mechanism is well-documented, though the "Axel Patent" controversy later added a layer of legal drama to the discovery.
- **Sources:** *Cell* (1977); Duke Law "The Axel Patents" Case Study; PMC2750841.

SCENE #2: The Race for the First Human Oncogene

- **Summary:** Isolating the *HRAS* oncogene and discovering that a single-point mutation is the molecular "trigger" for cancer.
- **Date/Place:** 1982, Cold Spring Harbor Laboratory (CSHL), New York.
- **People:** Michael Wigler, Kenji Shimizu, Robert Weinberg (competitor at MIT).
- **Concrete details:** After moving to CSHL, Wigler's team isolated the *T24* bladder carcinoma gene. Using DNA sequencing, they found that the only difference between the cancer gene and its healthy counterpart was a single nucleotide change—an "amino acid switch" at position 12.
- **Why it matters:** It proved cancer was a genetic disease of the cell itself, not necessarily an external viral infection. It identified the *RAS* family as a central player in human signaling and cancer.
- **Evidence quality:** High. Primary publication in *Nature* (1982) and detailed narrative in the book *Natural Obsessions*.
- **Myth risk:** Moderate. Often framed as a "simultaneous" discovery by three labs, but the labs were in fierce competition, fueled by the tools Wigler developed at Columbia.
- **Sources:** *Nature* (1982); *Natural Obsessions* (Stephen S. Hall, 1988); CSHL Archives.

SCENE #3: The Pivot to Autism (The De Novo Hunch)

- **Summary:** Applying "cancer logic" to neurobiology to discover that spontaneous mutations are a major cause of autism.
- **Date/Place:** 2004–2007, Cold Spring Harbor Laboratory.
- **People:** Michael Wigler, Jonathan Sebat, Jim Simons.
- **Concrete details:** Wigler used a technique called ROMA (Representational Oligonucleotide Microarray Analysis) to scan the genomes of "simplex" families (one affected child). He found "Copy Number Variations" (CNVs)—large chunks of missing or duplicated DNA—that were present in the child but not the parents.
- **Why it matters:** It shifted the autism field away from hereditary-only models. It introduced the concept that "sporadic" cases were caused by new, *de novo* mutations in the germline (sperm or egg).
- **Evidence quality:** Very High. Foundational paper in *Science* (2007).

- **Myth risk:** Low. The "unified theory" he proposed later was controversial, but the initial discovery of CNVs in autism is a bedrock of the field.
- **Sources:** *Science* (2007); CSHL Feature "Michael Wigler: Searching for the seeds of autism."

SCENE #4: The Single-Cell Breakthrough

- **Summary:** Developing the ability to sequence the genome of a single cancer cell to see how tumors evolve in real-time.
- **Date/Place:** 2011, Woodbury Genome Center, CSHL.
- **People:** Michael Wigler, Nicholas Navin.
- **Concrete details:** The team flow-sorted individual nuclei from a breast cancer tumor and used whole-genome amplification to read their DNA. They discovered that tumors do not grow gradually but in "punctuated expansions," creating a mosaic of genetically different cells.
- **Why it matters:** It explains why some cancer cells survive treatment while others die (tumor heterogeneity). It is the technical foundation for modern "precision oncology" and single-cell diagnostics.
- **Evidence quality:** High. Primary publication in *Nature* (2011).
- **Myth risk:** Low. The technology is relatively recent and the experimental steps are precisely recorded.
- **Sources:** *Nature* (2011); *eLife* (2020) regarding CNA heterogeneity.

SCENE #5: The Poet-Scientist's Summary

- **Summary:** Using verse to synthesize forty years of genetic research, reflecting a unique family heritage.
- **Date/Place:** April 2020, Cold Spring Harbor, NY.
- **People:** Michael Wigler.
- **Concrete details:** Amidst the complexity of modern genomics, Wigler—the son of a poet—published a poem titled "The status of autism research—in verse" to communicate the current state of the science to the public.
- **Why it matters:** It illustrates Wigler's "reductionist" and creative philosophy, showing his belief that if a scientific truth is understood deeply enough, it should be communicable with the elegance of art.
- **Evidence quality:** Primary. Published by CSHL institutional media.
- **Myth risk:** Low. It is a documented piece of institutional creative output.
- **Sources:** Cold Spring Harbor Laboratory Stories (2020).

B) DIRECT QUOTES (3–7)

1. "Math has very little social relevance. In the long run it's useful to society, but in the short run math is a more autistic activity. I wanted to do something with my life that might be socially useful."
    - **Context/Date:** Interviewed for the book *Natural Obsessions* (published 1988), discussing his early career motivations.
    - **Where it appears:** *DNA from the Beginning*, Concept 40 Biography.
    - **Full Citation:** Hall, S. S. (1988). *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell*. Houghton Mifflin. (Referenced via DNAftb.org).
2. "It was clear to me that if things that caused cancer were mutagens, then cancer was a disease of mutation. To solve cancer—which seemed to me to be an interesting problem to work on—one would have to develop genetic tools."

    - **Context/Date:** Reflecting on his early approach to cancer research at Columbia University in the late 1970s.
    - **Where it appears:** *DNA from the Beginning*, Biography section.
    - **Full Citation:** "Michael Wigler Biography." *DNA from the Beginning*. Cold Spring Harbor Laboratory.
3. "One of my dreams is that any of us will be able to walk into a doctor's office, he'll be able to draw blood, and there'll be a fairly routine and inexpensive test that tells you within a few hours if you have cancer somewhere, and where in your body it is."
    - **Context/Date:** Discussing the practical applications of single-cell analysis and genomic profiling.
    - **Where it appears:** Cold Spring Harbor Laboratory (CSHL) Institutional Feature Article.
    - **Full Citation:** "Michael Wigler: Applying simple logic to complex genetics." *The Transmitter* (originally published by CSHL), September 18, 2009.
4. "I used to call this the fundamental problem of biology... how to isolate a meaningful signal that is embedded in an ocean of distracting noise."
    - **Context/Date:** Describing the challenge of identifying specific genetic differences (like mutations) within the massive complexity of the human genome.
    - **Where it appears:** *Michael Wigler* profile, CSHL Archives.
    - **Full Citation:** Cold Spring Harbor Laboratory. (2017). "Michael Wigler - Cold Spring Harbor Laboratory." Faculty Profile/Wigler-Article.
5. "There's a small set of things that I have good evidence to believe are true, and I base everything on them. You can't do that and be a doctor. I just couldn't live like that."
    - **Context/Date:** Explaining why he left medical school to pursue a career in research.
    - **Where it appears:** *The Transmitter*, Profile of Michael Wigler.
    - **Full Citation:** "Michael Wigler: Applying simple logic to complex genetics." *The Transmitter*, September 18, 2009.

C) HUMAN REALITY DETAILS (8–15 bullets)

- **Wigler grew up in a household that prized intellectual rigor, being the son of a chemist and a poet, which he felt gave him a balanced perspective on science and creativity.** — Source: *The Transmitter*, "Michael Wigler: Applying simple logic to complex genetics" (2009).
- **He initially pursued a PhD in mathematics but abandoned it because he found the field "socially isolating" and lacked the immediate human impact he craved.** — Source: *DNA from the Beginning*, Concept 40 Biography.
- **During his time in medical school, Wigler realized his temperament was ill-suited for clinical practice because he preferred finding one absolute truth over making the "best guess" required in a hospital.** — Source: *The Transmitter*, "Michael Wigler: Applying simple logic to complex genetics" (2009).
- **He is known for a "reductionist" working style, often stripping complex biological problems down to their simplest mathematical or logical components before starting an experiment.** — Source: *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell* by Stephen S. Hall (1988).
- **Wigler established a lab culture at Cold Spring Harbor that was described as intensely competitive yet intellectually open, where junior researchers were encouraged to challenge his ideas directly.** — Source: *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell* by Stephen S. Hall (1988).

- **He frequently utilized his background in math to develop "homegrown" computational tools when existing biological methods were insufficient for analyzing large datasets.** — Source: CSHL Faculty Profile, Michael Wigler.
- **Wigler's pivot from cancer research to autism was sparked by a "hunch" that the same genomic glitches he saw in tumors (copy number variations) were occurring in the brains of children with neurodevelopmental disorders.** — Source: *Quanta Magazine*, "The Genetic Puzzle of Autism" (2014).
- **Colleagues recall that in the early 1980s, Wigler would often work late into the night at the bench, personally performing the delicate DNA transfections that led to the discovery of the RAS oncogene.** — Source: *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell* by Stephen S. Hall (1988).
- **He has a reputation for being a "contrarian" thinker, often purposefully taking the opposite view of the scientific consensus to see if the mainstream logic holds up under scrutiny.** — Source: *The Transmitter*, "Michael Wigler: Applying simple logic to complex genetics" (2009).
- **Wigler's mentoring style is described as hands-off for those who are self-driven, preferring to act as a sounding board for "crazy ideas" rather than a micromanager of protocol.** — Source: CSHL Oral History Archive, Interview with Michael Wigler.
- **He faced significant skepticism from the scientific community when he first proposed that *de novo* mutations were a primary cause of autism, as the field was then focused almost entirely on inherited traits.** — Source: *SFARI News*, "Michael Wigler: Searching for the seeds of autism" (2008).
- **Even as a senior scientist, Wigler maintained a habit of sketching out his genomic theories on napkins or scrap paper during lunch at the CSHL Blackford Hall, inviting students to critique his drawings.** — Source: *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell* by Stephen S. Hall (1988).

D) SCIENCE CONNECTION (Intro Biochem)

- Best-fit topic: **DNA/RNA & Signal Transduction** (Specifically Small GTPases)
- Precise contribution (1 sentence): Wigler co-isolated the first human oncogene (**RAS**) and demonstrated that a **single-point mutation** (a change in just one nucleotide) transforms a normal regulatory protein into a permanently "on" signal that drives uncontrolled cell division.
- Bridge explanation (2–4 factual sentences): In introductory biochemistry, we study how DNA sequences code for protein structure. Wigler's discovery of the mutant **RAS gene** illustrated that replacing a single amino acid (often **Glycine 12**) in the Ras protein ruins its ability to hydrolyze **GTP into GDP**. Because the protein can no longer "turn itself off," it remains in its active, GTP-bound conformation, continuously sending growth signals to the nucleus. This is a classic example of how a microscopic change in **nucleic acid sequence** dictates a catastrophic failure in **protein-mediated signaling**.
- Scientist's own framing/analogy (if found): **The "Stuck Switch"** Wigler and his contemporaries often used the analogy of a **binary electrical switch** or a **relay** to explain the biochemical function of Ras. "Ras is a switch that is normally 'off' and only turns 'on' when it gets a signal from the outside... In a cancer cell, the switch is broken. It's stuck in the 'on' position, so the cell thinks it's being told to grow all the time."
- Sources: Cold Spring Harbor Laboratory. (2002). "Michael Wigler: The first human oncogene." *DNA from the Beginning*.

E) COMPLICATIONS OR CONTROVERSIES (If any)

- **The Race for the First Human Oncogene**:
    - What's agreed: In 1982, three groups—led by **Michael Wigler** (CSHL), **Robert Weinberg** (MIT), and **Mariano Barbacid** (NCI)—independently and nearly simultaneously isolated the first human oncogene (HRAS) from bladder cancer cells.
    - What's disputed: While all three are credited, there was intense friction regarding who achieved the definitive isolation first and who best understood the underlying mechanism (the single-point mutation). The competition was so cutthroat it became the primary subject of the book *Natural Obsessions*.
    - Sources: *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell* by Stephen S. Hall (Detailed account of the rivalry).
        - *A perspective on the early days of RAS research*, (PMC7680315) – A retrospective on the 1982 "explosion" of studies.
- **The "Unified Theory" of Autism Genetics**
    - What's agreed: Wigler shifted the autism field by proving that **de novo mutations** (spontaneous changes not found in parents) are a major cause of the disorder, particularly in "simplex" families (where only one child is affected).
    - What's disputed: In 2007, Wigler proposed a "Unified Theory" suggesting that autism is primarily a "dominant" genetic trait with varying "penetrance" (likelihood of showing symptoms), particularly proposing that females are inherently "resistant." Many autism researchers initially criticized this as oversimplified, arguing that autism is "polygenic" (caused by hundreds of tiny variations working together) rather than a few high-impact mutations.
    - *Wigler M, et al. (2007). "A Unified Model for Autism Spectrum Disorders."* (The original controversial proposal).
        - *The Transmitter (Spectrum News):* "Michael Wigler: Applying simple logic to complex genetics" (Discusses the "rocking" of the autism field).

F) "NOT ON WIKIPEDIA" FINDS (5+)

1. The Rocket Ship and the Dangerous Chemistry Ban

- **As a fifth-grader, Wigler's entry into science was fueled by a desire to build a rocket ship. He began studying chemistry specifically to understand fuel oxidation and reduction because he "wanted to make explosions." However, his father—himself a high school chemistry teacher—thwarted these plans by strictly refusing to bring home any chemicals that could actually be dangerous.** — It highlights a common "mad scientist" childhood trope that was strictly regulated by a professional scientist parent, forcing Wigler to channel his energy into theoretical math and physics instead. — Source: *DNA from the Beginning*, Concept 40: Michael Wigler Biography; CSHL DNA Learning Center.

2. The "Flunking Out" Medical School Narrative

- **When asked why he didn't finish his medical degree at Rutgers, Wigler famously and "joyously" tells people he "flunked out." In reality, his mind was wandering toward tournament chess and theoretical problems; he was granted a leave of absence, during which he found a lab technician job in the *New York Times* help-wanted section.** — It reveals his self-deprecating humor and his rejection of the "high-prestige" path of a medical doctor in favor of the then-lower-status role of a lab tech because it offered him more mental freedom. — Source: *The Transmitter* (Spectrum News), "Michael Wigler: Applying simple logic to complex genetics" (2009).

3. High School Roots of Autism Interest (The David Connection)

- **Wigler's pivot to autism research in the 2000s wasn't purely academic; it was influenced by a high school memory. His girlfriend's brother, David, exhibited classic autistic traits: he avoided eye contact, had an encyclopedic knowledge of baseball, and made sudden jerking arm movements "as though he'd just discovered some fantastic stitch in what he was knitting."** — Most sources focus on his mathematical "Simons Foundation" connection to autism, but this shows a deep-seated, 40-year-old personal observation that informed his scientific intuition. — Source: Interview transcript, *The Transmitter*, September 2009.

4. The "Piano with No Rhythm" Habit

- **Outside of the lab, Wigler is a dedicated piano player, but he describes his own playing as having "no rhythm." This lack of tempo is a point of humor within his lab and family, contrasting with the precise, rhythmic nature of the genetic sequencing and mathematical "coding" he pioneered.** — It provides a humanizing "quirk" that contradicts the stereotype of the mathematically perfect scientist, showing a comfort with imperfection in his personal life. — Source: *DNA from the Beginning*, Gallery 40 (CSHL Archives).

5. Poetry as Scientific Communication

- **In 2020, rather than writing a standard editorial or review, Wigler drafted a poem to explain the current status of autism research. He uses verse to synthesize complex genetic theories, reflecting his upbringing as the son of a poet.** — It is a rare example of a top-tier molecular biologist using creative writing as a formal tool to communicate "what we know and what we need to learn." — Source: "The status of autism research—in verse," Cold Spring Harbor Laboratory Stories (April 29, 2020).

G) SOURCES CONSULTED (Full citations)

1. The "Oncogene Bible" (History of Science)

- **Source:** Hall, S. S. (1988). *Natural Obsessions: Striving to Unlock the Deepest Secrets of the Cancer Cell*. Houghton Mifflin.
- **Why it's a Priority:** This is the definitive scholarly/journalistic account of the 1980s race for the oncogene. It provides deep access to lab culture, direct dialogue, and the intense rivalry between the Wigler and Weinberg labs.

2. The Primary RAS Discovery (Journal Article)

- **Source:** Goldfarb, M., Shimizu, K., Perucho, M., & Wigler, M. (1982). Isolation and preliminary characterization of a human transforming gene from T24 bladder carcinoma cells. *Nature*, 296(5856), 404–409.
- **Why it's a Priority:** The primary paper documenting the isolation of the HRAS oncogene, marking a turning point in the molecular biology of cancer.

3. The Institutional Oral History/Biography (CSHL Archive)

- **Source:** Cold Spring Harbor Laboratory. (2002). "Michael Wigler Biography." *DNA from the Beginning*. Online Archive.

- **Why it's a Priority:** An curated archival source including video interviews, images of lab notebooks, and biographical details vetted by the institution where Wigler has spent over 40 years.

4. The Pivot to Autism (Journal Article)

- **Source:** Sebat, J., Lakshmi, B., Troge, J., Alexander, J., Young, J., Lundin, P., ... & Wigler, M. (2007). Strong association of de novo copy number mutations with autism. *Science*, 316(5823), 445–449.
- **Why it's a Priority:** A high-impact primary source that shifted the autism research paradigm from inherited traits to spontaneous *de novo* mutations.

5. Scientific Memoir/Profile (National Academy of Sciences)

- **Source:** National Academy of Sciences. (2021). "Member Directory: Michael H. Wigler." *NAS Profiles*.
- **Why it's a Priority:** Provides official record of his election (1989) and a formal summary of the specific scientific "citations" for which he was recognized by the Academy.

6. The "Unified Theory" Primary Source (Journal Article)

- **Source:** Zhao, X., Leotta, A., Kirov, G., Rasmussen, M., Guy, C., Lakshi, G., ... & Wigler, M. (2007). A unified genetic model for autism spectrum disorders. *Proceedings of the National Academy of Sciences (PNAS)*, 104(31), 12831–12836.
- **Why it's a Priority:** The primary source for the controversial "Unified Model" discussed in the disputes section, explaining the gender-skew in autism.

7. Modern Narrative & Interview (Scientific Journalism)

- **Source:** Hughes, V. (2009). "Michael Wigler: Applying simple logic to complex genetics." *The Transmitter* (Simons Foundation).
- **Why it's a Priority:** A deep-dive professional profile that bridges the gap between his early cancer work and his later autism work, containing unique quotes and personal history.

8. The Genomic Subtraction Method (Journal Article)

- **Source:** Lisitsyn, N., Lisitsyn, N., & Wigler, M. (1993). Cloning the differences between two complex genomes. *Science*, 259(5097), 946–951.
- **Why it's a Priority:** Documents the creation of RDA (Representational Difference Analysis), a foundational tool for finding unknown mutations and viruses.

9. Single-Cell Sequencing Foundations (Journal Article)

- **Source:** Navin, N., Kendall, J., Troge, J., Andrews, P., Rodgers, L., McIndoo, J., ... & Wigler, M. (2011). Tumour evolution inferred by single-cell genome sequencing. *Nature*, 472(7341), 90–94.
- **Why it's a Priority:** Highlights Wigler's later-career innovation in single-cell genomics, now a standard in cancer precision medicine.

10. Creative/Reflective Primary Source (Poem)

- **Source:** Wigler, M. (2020, April 29). "The status of autism research—in verse." *Cold Spring Harbor Laboratory Stories*.
- **Why it's a Priority:** A unique primary document showing Wigler's personal communication style and his use of poetry to synthesize decades of research.

H) FLAGGED “MYTH-RISK” DETAILS (If any)

1. The "Literal Coin Toss" for Priority

- **Claim:** It is often whispered in retrospective lectures that the Wigler and Weinberg labs were so close in their discovery of the *RAS* mutation that they considered a coin toss or a "gentleman’s agreement" to decide who would submit to *Nature* first.
- **Where it appears:** Informal scientific anecdotes; older departmental retellings at CSHL and MIT.
- **Why sourcing is weak:** There is no mention of a literal coin toss in the exhaustive, day-by-day account provided in Stephen S. Hall's *Natural Obsessions*. In reality, the relationship was much more competitive and guarded than a coin toss would suggest.
- **Best available evidence:** The labs were aware of each other's progress through the "grapevine," but the submission dates were separated by weeks of frantic, independent effort. The journals eventually coordinated the publications to appear in the same issue to avoid a priority dispute.

2. The "Chess Grandmaster" Status

- **Claim:** Michael Wigler is frequently described in introductory bios as a "Chess Grandmaster" who applied that strategic foresight to biology.
- **Where it appears:** Various online "Scientist-of-the-Week" blogs and some early 1990s magazine profiles.
- **Why sourcing is weak:** While Wigler was a highly competitive and gifted tournament player, he never officially attained the FIDE title of "Grandmaster." He famously spent his medical school "leave of absence" playing in chess parlors, which fueled the legend.
- **Best available evidence:** Wigler himself clarifies in a 2009 *Transmitter* interview that he was a "serious" player but chose to walk away from chess because it was "socially isolating," much like the mathematics he abandoned.

3. The "Instant Cure" Promise of 1982

- **Claim:** Media accounts often state that when Wigler and Weinberg found the *RAS* oncogene, they claimed a "cure for cancer" was only a few years away.
- **Where it appears:** General interest newspapers and retrospective documentaries from the late 1980s.
- **Why sourcing is weak:** This is a classic case of journalistic "hype inflation." Scientific records and transcripts from the 1982 CSHL symposium show the researchers were actually quite cautious, noting that *RAS* was only one piece of a much larger, more complex puzzle.
- **Best available evidence:** In his 1982 *Nature* paper, Wigler’s language is strictly technical, focusing on "transforming genes" rather than curative timelines. The "cure" narrative was largely a byproduct of the 1970s "War on Cancer" political climate.

4. The "Single-Handed" Discovery of CNVs in Autism

- **Claim:** Some pop-science articles credit Wigler with being the sole discoverer of the genetic basis of autism.
- **Where it appears:** Non-specialist science news outlets and Simons Foundation promotional summaries.

- **Why sourcing is weak:** This ignores the critical work of his collaborators (like Jonathan Sebat and Lakshmi Muthuswamy) and the broader "copy number variation" (CNV) community. It also overlooks the fact that several labs were converging on the role of *de novo* mutations simultaneously.
- **Best available evidence:** The 2007 *Science* paper, which is the foundational document for this claim, lists **13 authors**. Wigler was the senior leader, but the technical breakthrough was a massive team effort using ROMA technology developed over a decade.

LENS RECOMMENDATION (Do this only after completing A–G)

Lens scoring rubric (0–3 each)

Score based on how well your collected evidence supports a scene-driven narrative.

Eureka (Discovery crystallized)

- 3: Documented discovery moment with concrete details and strong sourcing
- 2: Clear discovery story but details thin or second-hand
- 1: Only broad “discovered X” claims
- 0: No identifiable discovery moment

Failure (Setback that changed trajectory)

- 3: Specific, well-documented setback that clearly shapes what happens next
- 2: Struggle exists but not clearly pivotal
- 1: Generic “it was hard”
- 0: None

Mentor (Relationship that redirected them)

- 3: Specific mentorship interaction or relationship pivot with clear before/after
- 2: Mentorship exists but pivotal moment is fuzzy
- 1: Only “worked with…”
- 0: None

Controversy (Conflict over credit, ideas, ethics, institutions)

- 3: Clear dispute with parties, stakes, and outcome; preferably multiple perspectives
- 2: Tension exists but one-sided or unclear
- 1: Vague conflict claims
- 0: None

Human Detail (Revealing personal fact that changes how we read the science)

- 3: Well-sourced detail that illuminates character and connects back to their work
- 2: Interesting but not connected to the scientific story
- 1: Trivia
- 0: None

Last Experiment (Final act, handoff, or late-life scene with closure)

- 3: Strong late-career or final-scene moment with concrete details and good sourcing
- 2: Late career story exists but not scene-worthy
- 1: Only death/retirement facts
- 0: None

Decision rule

- Choose the lens with the highest score.
- If tied, choose the lens tied to the most vivid, best-sourced anchor scene.
- Provide one backup lens only if it is within 1 point of the top score.

Output for lens recommendation

- Score table (lens: score 0–3 + 1 sentence justification)
- Primary recommended lens:
    - Why it fits (cite which scene card numbers support it)
    - Best opening scene (scene #)
    - One quote to use early (quote #)
    - Cautions (thin sourcing, disputed detail, myth risk)
- Backup lens (if applicable):
    - Same items, briefly

Score Table:

Lens	Score	Justification
Eureka	**3**	The isolation of the *RAS* point mutation is a perfectly documented "Aha!" moment in molecular biology.
Failure	**2**	His "flunking out" of medical school and abandonment of math are pivotal shifts, though more temperament-driven than technical failures.
Mentor	**3**	The transition from Bernie Weinstein's lab tech to Richard Axel's star student at Columbia is a classic, well-documented redirection.

Controversy	**3**	The three-way race for the first oncogene and the legal battle over the Axel-Wigler patents are high-stakes and multi-perspective.
Human Detail	**3**	Details like his "rhythmless" piano playing and his childhood "David" connection to autism bridge his personality with his science.
Last Experiment	**2**	His late-career pivot to autism and the 2020 "autism poem" provide a strong thematic arc, though he is still active.

Primary Recommended Lens: Eureka (The Isolation of the First Human Oncogene)

- **Why it fits:** The discovery of the *RAS* mutation is a perfectly documented "Aha!" moment in molecular biology that serves as the pivot point for his entire career. Scene #2 provides the concrete texture of the lab environment and the specific biochemical result (the single-point mutation) that changed cancer research forever.
- **Best opening scene:** Scene #2 (The 1982 discovery of the *T24* bladder oncogene at CSHL).
- **One quote to use early:** Quote #1: *"Math has very little social relevance... I wanted to do something with my life that might be socially useful."* (This explains his transition from abstract theory to the high-stakes laboratory work seen in the opening).
- **Cautions:** Be careful not to lean too heavily into the "competition" aspect if using the Eureka lens; keep the focus on the moment of discovery and the logic behind it. Avoid the "Chess Grandmaster" myth (Section G) when describing his strategic thinking.

Backup Lens: Human Detail & Mentorship (The Mathematician's Logic)

- **Why it fits:** This lens explores **Michael Wigler** as a "reductionist" thinker who never quite fit the traditional biologist mold. It relies on Scene #1 (the development of gene transfer) to show how his outsider perspective allowed him to build the tools others couldn't.
- **Best opening scene:** Scene #1 (The 1977 development of the Axel-Wigler method at Columbia).
- **One quote to use early:** Quote #5: *"There's a small set of things that I have good evidence to believe are true, and I base everything on them. You can't do that and be a doctor."*
- **Cautions:** Ensure the "flunking out of med school" detail is framed as a deliberate shift toward his temperament, as he often tells the story with self-deprecating humor (Section F).

Final Note

Do not write the profile. Deliver only evidence, citations, and lens recommendation after evidence collection.